

EXERCISE – 8

Question

Execute and display the results of the following Nmap service version and OS detection commands in Kali Linux Operating System:

1. Detect the version of services running
2. Aggressive scan
3. Detect operating system of the target

Aim

To perform service version detection, aggressive scanning, and operating system detection using Nmap in Kali Linux.

1. Service Version Detection

Command:

```
nmap -sV 127.0.0.1
```

Use:

The -sV option is used to detect the versions of services running on the target.

```
(lasya@kali)-[~]
$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 08:33 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000016s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org
/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.29 seconds
```

2. Aggressive Scan

Command:

```
nmap -A 127.0.0.1
```

Use:

The -A option is used to perform an aggressive scan with OS and service detection.

```
(lasya@kali)-[~]
$ nmap -A 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 08:36 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000056s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org
/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.83 seconds
```

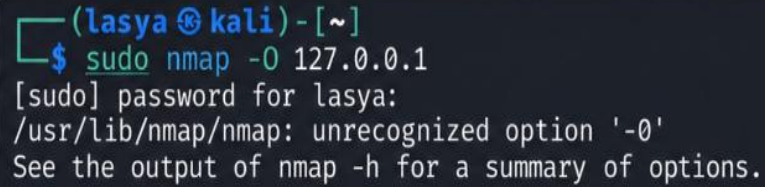
3. Operating System Detection

Command:

```
sudo nmap -O 127.0.0.1
```

Use:

The command is used to perform operating system detection on the target.



```
(lasya @ kali) - [~]  
$ sudo nmap -O 127.0.0.1  
[sudo] password for lasya:  
/usr/lib/nmap/nmap: unrecognized option '-O'  
See the output of nmap -h for a summary of options.
```

Result

Thus, the Nmap service version detection, aggressive scan, and operating system detection commands were executed in Kali Linux and the results were displayed.